

Airbus Product Standard

Special S

Extruded sections

Aluminium alloy (7175)

Thickness $1,3 \text{ mm} \leq a \leq 11,0 \text{ mm}$

Uncontrolled copy when printed (: BERNINCHES FINIADO, MARINA [ES], 22/12/2022)

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1 Scope

This standard specifies the dimensions, tolerances and mass of extruded sections in aluminium alloy 7175 with peripheral coarse grain control, $1,3 \text{ mm} \leq a \leq 11,0 \text{ mm}$, for aerospace applications.

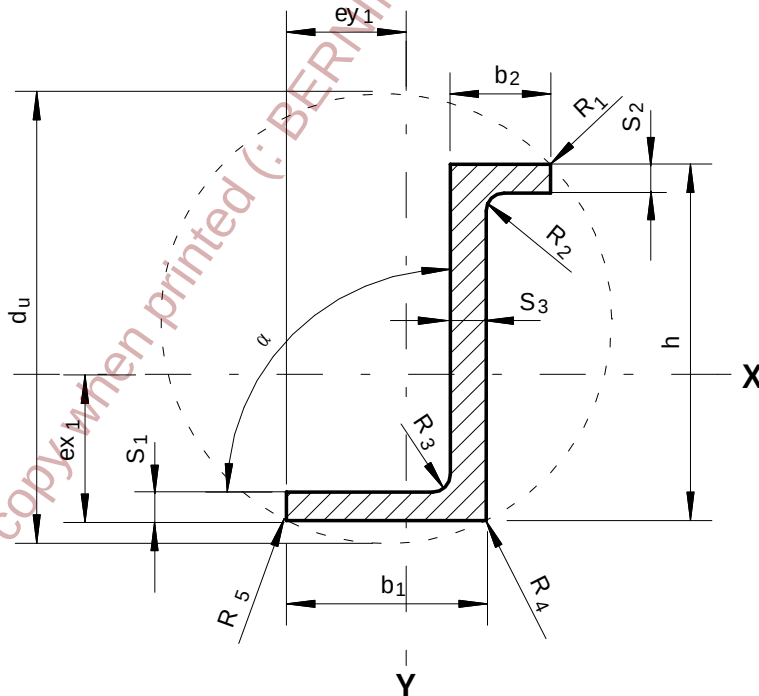
2 Normative References

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS03-05-000	Technical Specification – Aluminium and aluminium alloy – Wrought products – Bar and Section
AIMS03-05-009	Material Specification – Aluminium alloy (7175) – Solution treated, controlled stretched and artificially aged (T73511). Extruded section $1,2 \text{ mm} \leq a \leq 38,0 \text{ mm}$ with peripheral coarse grain control
AIMS03-05-012	Material Specification – Aluminium alloy (7175). Solution treated, controlled stretched and artificially aged (T79511). Extruded section – $1,2 \text{ mm} \leq a \leq 38,0 \text{ mm}$ with peripheral coarse grain control
AIMS03-05-019	Material Specification – Aluminium alloy (7175) – Solution treated and artificially aged (T73) – Extruded section – $1,2 \text{ mm} \leq a \leq 38,0 \text{ mm}$ with peripheral coarse grain control
DIN1748 Part 4	Aluminum and wrought aluminum alloy extruded profiles - Permissible Deviations
EN2066	Extruded section in aluminum and aluminum alloys, general tolerances

3 Form

See Figure 1



(unspecified radi 0,5 mm typ.)
 (for size codes 143, 160, 173, 183, 192, 208, 245 and 284
 unspecified radi 0,2 mm typ.)
 (for size code 1094 unspecified radi 1,5 mm typ.)

Figure 1: Form

4 Recommended Dimensions, Mass and Static Values

4.1 Dimensions, cross-sections and mass

See Table 1

Table 1: Dimensions, cross-sections and mass

Dimensions in mm

Size Code	h	b ₁	b ₂	S ₁	S ₂	S ₃	R ₁	R ₂	R ₃	R ₄	R ₅	α	d _u	d _u ----- a _{min}	Cross-section mm ²	Length	Linear mass ^{a)} kg/m
64	20,7		4,0		1,2								38,85	32,0	64,2		0,18
76	21,5				2,0								44,51	37,0	75,5		0,21
88	23,0	30,0	8,0	1,2	3,5	1,2	0,4	2,0	2,0	0,4	0,4		45,03	38,0	87,5		0,25
92	23,5				4,0								45,27		91,5		0,26
109	24,0		11,0		4,5								50,09	42,0	109,0		0,30
135	25,0	24,0	7,0	2,5	4,0	2,5	-			2,0	-		39,0	15,6	135,0		0,38
130	25,0	30,0	12,0	1,4	5,0	1,4	0,4	2,0	2,0	0,4	0,4		51,65	37,0	129,5		0,36
84	25,4	23,0	8,0	1,6	2,0	1,3	-	2,5	2,5	-	-		39,0	30,0	83,9		0,23
115	27,0	35,0	10,0	1,6	2,2	1,6	0,2	1,6	1,6	1,8	0,2		51,1	31,9	115,4		0,32
185		22,0	6,0	3,6	6,9	3,2	2,0	1,6	2,0	1,6	0,5		36,5	11,4	185,2		0,52
255	28,0	28,0	15,0	4,0	4,0	4,0	-	2,0	4,0	-	-		51,0	12,8	255,0		0,71
144	28,8	24,0	8,5	2,2	5,0	2,2	1,0	2,0	2,0	1,0	0,5	90°	42,3	19,2	143,6		0,40
101	29,0	31,6	6,0	1,6	1,6	1,6		1,6	1,6	1,6	-		46,2	28,8	101,0	stock length not less than 2000 and exact length by agreement	0,28
107		22,0	5,0	2,0	2,0	2,0		2,0	2,0	0,8	0,8		39,2	19,6	107,0		0,30
192		25,0	8,5	3,0	6,0	3,0		2,5	2,5	-			42,8	14,3	192,0		0,55
225		26,0	10,0	3,5	6,0	3,5		2,5	2,5	-			44,2	12,7	225,4		0,64
137	30,0	32,0	6,0	2,0	3,0	2,0		2,5	2,5	0,8			46,9	23,5	134,7		0,37
175		24,0	24,0	2,0	3,0	2,0		2,5	2,5				54,9	27,5	172,7		0,48
258		22,8	9,0	7,2	4,5	2,8		2,5	2,5	-			41,7	14,9	258,6		0,72
288		34,0	15,0	4,0	4,0	4,0		2,0	4,0				53,7	13,4	288,0		0,81
143	31,0	35,0	6,0	2,0	3,0	2,0	0,2	2,5	2,5	0,2	0,2		49,8	24,9	143,0		0,40
160				2,5											160,0		0,44
453	32,0	11,0	25,0	4,5	11	7,0	-	3,0	3,0	-	-		43,7	9,7	453,0		1,27
170							-			-	-	92,4°	52,47	20,9	170,2		0,47
172		26,0	12,0	2,5	2,5	2,5	-	2,5	2,5	-	-	94,9°	54,08	21,6	170,2		0,47
174							-			-	-	96,8°	55,39	22,2	170,2		0,47
173	34,0		8,0	2,2	4,0	2,2							53,1	24,1	173,0		0,48
183															183,4		0,51
208		35,0	10,0	2,5	5,0	2,5	0,2	2,5	2,5	0,2	0,2	90°	53,4	21,7	207,9		0,58
245			11,0	3,0	5,5	3,0							54,8	18,2	245,0		0,68
284			12,0	3,5		3,5							55,2	15,7	283,7		0,79

(continued)

Table 1: Dimensions, cross-sections and mass (concluded)

Dimensions in mm

Size Code	h	b ₁	b ₂	S ₁	S ₂	S ₃	R ₁	R ₂	R ₃	R ₄	R ₅	α	d _u	d _u ---- a _{min}	Cross-section mm ²	Length	Linear mass ^{a)} kg/m
153	35,0	28,0	15,0	1,8	4,0	1,4	0,4	2,0	2,0	0,4	0,4	90°	58,1	32,0	152,8	stock length not less than 2000 and exact length by agree-ment	0,43
159	35,0	28,0	15,0	1,8		1,6	0,4	2,0	2,0	0,4	0,4		57,81	32,0	158,6		0,44
323	38,0	28,0	10,0	4,5	7,0	4,5	1,0	4,5	4,5	-	-		50,0	11,1	323,0		0,91
356		34,0	15,0	4,0		4,0	-	4,0	4,0	-	-		58,5	14,6	356,0		1,00
176	40,0	28,0	15,0	2,0	4,0	1,6	0,4	2,0	2,0	0,4	0,4		60,44	30,0	171,9		0,48
244	45,0	30,0	10,0	3,0	3,0	3,0	-	4,0	4,0	-	-		57,9	19,3	244,0		0,68
171	50,0	22,0	26,0	1,8	1,8	1,8	0,6	1,8	1,8	0,6	0,6		67,6	37,6	171,0		0,49
161		24,0	15,0	1,6	3,5	1,4	0,4	2,0	2,0	0,4	0,4		68,41	49,0	159,8		0,45
301	62,0	65,0	24,5	2,0	2,0	2,0	-	4,0	4,0	-	-		107,2	53,6	301,8		0,86
1094	187	27,0	38,0	9,0	9,0	3,0	1,5	4,5	4,5	1,5	1,5		196,2	65,4	1094,0		3,06
1812		28,5	40,5			7,0							200,1	28,6	1812,0		5,07
1726		27,0	45,0	10	10	6,0		4,0	4,0	1,5	1,5		197,7	32,9	1726,0		4,83
a) For information, calculated with a density of 2,8 kg/dm ³																	

4.2 Static Values

See Table 2

Table 2: Static Values

Size code	ex1 mm	ey1 mm	I _x mm ⁴	I _y mm ⁴
64	5,62	8,53	1 555	7 692
76	7,84	6,73	1 859	11 014
84	9,64	18,23	11 092	1 783
88	9,83	5,42	2 126	13 898
92	10,41	5,06	2 204	14 789
101	9,80	23,57	10 489	10 308
107	10,60	17,10	11 600	4 500
109	12,31	3,12	2 316	19 027
115	9,70	26,9	12 442	15 762
130	12,90	2,80	2 813	23 565
135	9,50	18,40	10 800	7 500
137	9,97	24,14	23 950	5 185
143	9,97	26,16	6 325	28 575
144	12,6	19,8	21 976	3 016
153	1,00	17,81	4 357	42 874
159	0,85	17,78	4 509	42 952
160	9,17	25,17	7 043	31 620
161	25,78	1,52	4 142	71 324
170	13,74	21,66	27 262	12 593
171	26,00	22,00	65 500	14 800
172	13,68	22,24	26 983	13 883
173	12,36	27,14	40 772	7 907
174	13,63	22,68	26 702	14 930
175	16,59	24,53	25 166	19 900
176	1,16	19,76	5 335	56 631
183	11,82	26,56	43 103	8 447
185	10,41	16,48	15 788	7 724
192	12,40	19,94	29 537	4 521
208	13,42	27,77	51 185	8 824
225	12,40	20,60	26 283	14 818
244	17,30	24,00	63 600	18 800
245	13,46	27,62	59 963	10 240
255	11,50	22,00	26 200	22 100
258	10,04	15,54	24 064	15 346
284	13,35	27,33	67 180	11 589

(continued)

Table 2: Static Values (Concluded)

Size code	ex1 mm	ey1 mm	I _x mm ⁴	I _y mm ⁴
288	11,50	26,00	22 800	36 400
301	23,00	52,30	179 531	141 873
323	15,30	21,80	56 600	21 800
356	16,60	27,90	70 900	42 900
453	20,10	12,90	39 200	25 100
1094	101,50	28,30	5 947 792	84 364
1726	102,74	27,43	7 863 571	274 695
1812	98,8	26,8	7 748 180	197 920

5 Tolerances

5.1 Dimensional Tolerances

See Table 3

5.2 Geometrical Tolerances

See Table 3

Table 3: Geometrical Tolerances

Standard	Tolerance Class	Code	
EN2066	Standard	E	Use only for order designation
DIN1748 Part 4	Standard	N	
	Precision	F	If this class is required, the code letter F shall be used for standard designation and order designation

6 Material

See Table 4

Table 4: Material

Thickness range	Material					
	delivered			use		
	AIMS spec. No.	Condition of treatment	Code	AIMS spec. No.	Condition of treatment	Code
2,0 mm ≤ a ≤ 28,6 mm	03-05-009	T73511	A	03-05-009	T73511	A
	03-05-019	O	E	03-05-019	T73	B
		F	F			
	03-05-012	T79511	C	03-05-012	T79511	C

7 Designation

This type of standard shall be designated according to the philosophy of the following example:

	Description block	Identity block			
	Section	ABS5059	A	244	F
Number of this standard					
Material code (see table 4)					
Size code (see table 1)					
If required use tolerance code F (see table 3)					

8 Technical Specification

AIMS03-05-000

Record of Revisions

Issue	Clause modified	Description of Modification
1 06.2000	-	New Standard
2 09.2002	-	Size code 101 added (Airbus-F request).
3 12.2002	-	Size code 084; 160; 183; 208; 284 added (Airbus-F request). In table 4 the condition O and F temper added (Airbus-F request).
4 03.2003	-	Size codes 143, 173, and 245 added (Airbus-F request). Size code 084 modified to 84 (Airbus-F request).
5 09.2003	-	Size code 192 and 1094 added.
6 04.2005	-	Size code 137 added. Static values lx and ly for size code 101 corrected. (Was lx = 17215 and ly = 3796 ; Corrected lx = 10489 and ly = 10308)
7 02.2006	-	In figure 1 and table 1 dimension α added. Size code 170, 172 and 174 added. AIMS 03-05-020 reference cancelled.
8 04.2006	-	Size code 185 added.
9 07.2006	-	Size code 1726 added.
10 08.2011	-	Size codes 1812 and 115 added (A380 request). Static values for size codes 115 and 1812 added. In title and chapter 2 the dimensional range corrected from $1,2\text{mm} \leq a \leq 38,0\text{mm}$ to $1,3\text{mm} \leq a \leq 11,0\text{mm}$
11 11.2006	-	Size code 175 and 258 added.
12 03.2021	-	Size codes 64, 76, 88, 92, 109, 130, 159, 161 and 176 added.
13 11.2021	-	Size code 144 has been added.